

Name: _____

Date: _____

MULTIPLICATION, (LONG MULTIPLICATION)

LO: I can multiply 2- and 3-digit numbers using the long multiplication (column) method.

What is long multiplication?

Long multiplication is the column method for **multi-digit** × multi-digit. We multiply by each digit separately, then add.

Place-value rule: when multiplying by tens, put a 0 in the units column first. By hundreds, put 00.

Examples

●	$23 \times 45 = 1035$	2-digit
●	$146 \times 27 = 3942$	3-by-2 digit
●	Multiply units, then tens	two rows then add
●	Forget the placeholder 0	answer wrong
●	Multiply only the units	incomplete

Key idea

Multiply by each digit of the bottom number separately. Each new row shifts left by one column (place value).

$$\textcircled{23} \times \textcircled{45} = \textcircled{1035}$$

Two rows, then add

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – 2 by 2 digit

Calculate 23×45

$$23 \times 5 = 115 \text{ (units row)}$$

$$23 \times 40 = 920 \text{ (tens row, with 0 placeholder)}$$

$$115 + 920 = 1035$$

$$= 1035$$

Always add a placeholder 0 when multiplying by the tens digit.

Example 2 – with carrying

Calculate 47×36

$$47 \times 6 = 282 \text{ (units row)}$$

$$47 \times 30 = 1410 \text{ (tens row)}$$

$$282 + 1410 = 1692$$

$$= 1692$$

Show carries clearly above each column.

Example 3 – 3 by 2 digit

Calculate 146×27

$$146 \times 7 = 1022$$

$$146 \times 20 = 2920$$

$$1022 + 2920 = 3942$$

$$= 3942$$

3-digit numbers follow the same rules; just more digits.

Example 4 – with zeros

Calculate 304×25

$$304 \times 5 = 1520$$

$$304 \times 20 = 6080$$

$$1520 + 6080 = 7600$$

$$= 7600$$

The zero in 304 means $\times 0$ gives 0 in that column.

YOUR TURN – PRACTICE (deliberate practice)

1. 2 by 1 digit

Use column method.

- a) 34×7
- b) 56×8
- c) 73×9
- d) 85×6

2. 2 by 2 digit

Two rows, then add. Don't forget the placeholder 0.

- a) 23×45
- b) 62×38
- c) 47×26
- d) 84×19

3. 3 by 2 digit

Same method, more digits.

- a) 146×27
- b) 234×35
- c) 508×14
- d) 672×25

4. Word problems

Set up a multiplication, solve.

- a) 27 boxes of 36 = ____
- b) School trip: 14 coaches $\times$ 52 seats = ____
- c) 8 weeks at 175 mins/week = ____
- d) 23 packs of 48 stickers = ____

5. Mixed

Use the column method.

- a) 39×46
- b) 125×16
- c) 78×53
- d) 246×18
- e) 100×100

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $23 \times 45 = 23 \times 4 \times 5$ ☐

Correct answer: _____

Reason: _____

b) Add a placeholder 0 when multiplying by tens ☐

Correct answer: _____

Reason: _____

c) $47 \times 36 = 1692$ ☐

Correct answer: _____

Reason: _____

d) Multiplication is commutative: $23 \times 45 = 45 \times 23$ ☐

Correct answer: _____

Reason: _____

e) $304 \times 25 = 760$ (forget last 0) ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A theatre has **27 rows** with **48 seats** per row. How many seats in total? If **£12.50** per ticket, what's the maximum revenue per show?

Expression: _____

Simplified: _____

b) A factory packs **148 chocolates per box**, **36 boxes per crate**, and ships **12 crates a day**. How many chocolates ship per day? Show your working clearly.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

